

From a Vibrating String to the Uncertainty Principle

First-Principles Derivation Series · 04 — Fourier Modes and Physical States

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From a Vibrating String to the Uncertainty Principle

Author's note — on this series and how this post is written

This is a series that re-derives physics from first principles, with the floor set at undergraduate engineering physics: Newtonian mechanics, energy conservation, Fourier, ordinary differential equations. Going below that (say, deriving the Einstein field equations from an action principle) is a different rabbit hole and not on this chain.

The derivation in this Post 04 is a genuine back-and-forth: one side drives the derivation, the other only judges “does this close, or not.” I’ve kept that process as intact as possible — including the places where things got **stuck**, and how they got unstuck. Because the hard part was never the final formula; it was the handful of counter-intuitive **cognitive checkpoints** along the way. I mark those with Checkpoint.

One concept runs through the whole post: the **IOU**. Some pieces of rigor (a certain integral, a convergence, a postulate) we **explicitly do not prove** — we record a debt and keep it hanging, rather than pretending it’s settled. First principles does *not* mean welding the rebar of every foundation slab — it means **knowing why each step holds, and honestly marking which pieces are black boxes**. There’s a full IOU ledger at the end.

0. Where this block sits in the whole chain

This block (Fourier modes) is a hinge, connecting on both ends:

- **Backward:** the previous block derived interference / probability amplitudes — the “superposition of waves.” This block answers a more basic question: **why can we decompose into “modes,” and why are modes special?** The next block (05, Schrödinger) is, at bottom, just “swap in a different dispersion relation.”
- **Forward:** this block’s conclusions **are literally the machine for the later cosmology ML project**. Decomposing the density field by wavenumber k , each k -mode evolving independently in linear theory, describing statistics with the power spectrum $P(k)$ — it’s the same machinery, word for word.

The block lands on three results:

1. **The 1D wave equation** $\partial_{tt}y = c^2\partial_{xx}y$ (from Newtonian mechanics)
2. **Mode decomposition + complete orthogonal basis:** any wave = a superposition of sinusoidal modes; a function is a vector in an infinite-dimensional space
3. **The uncertainty principle** $\Delta x \cdot \Delta k \geq \frac{1}{2}$ (a pure Fourier theorem, nothing quantum; 05 adds one line, $p = \hbar k$, to make it Heisenberg)

1. Deriving the 1D wave equation from Newtonian mechanics

1.1 Modeling: put every assumption on the table

A uniform, flexible string, stretched taut along the x -axis at rest, tension magnitude T , linear density μ (mass per unit length). Give it a transverse disturbance; let $y(x, t)$ be the transverse displacement of the element at x from equilibrium.

Assumptions, laid out one by one (each one is a tab we’ll settle later):

#	Assumption	Role
A1	Perfectly flexible (no bending stiffness)	Internal force can only lie along the tangent — this fixes the direction of the force
A2	Purely transverse motion (elements move only up/down, not along x)	$v_x = 0$
A3	Small slope $ \partial y / \partial x \ll 1$	The one and only approximation; linearity comes from here
A4	Tension T is the same everywhere	(!) used as an assumption first, but it’s actually a consequence of A2+A3; audited later
A5	No gravity, no damping	For cleanliness

Terminology checkpoint: transverse vs. longitudinal

The reference is not the screen — it’s the **direction of wave propagation**. The string is stretched along x and the wave travels along x , so x is the “longitudinal” axis; the displacement is along y , **perpendicular to propagation**, so it’s called **transverse**. In everyday language “longitudinal = vertical, transverse = horizontal,” which is exactly backwards from this convention on both counts — so the instinct “the disturbance should be called longitudinal” is natural; the physics convention just slices by “relative to propagation,” not by which way is vertical on the page.

1.2 Force analysis: tension along the tangent, both ends pulling outward

Take the element $[x, x + dx]$. By A1, the internal force at each end lies along the tangent, magnitude T , at angles $\theta(x)$ and $\theta(x + dx)$ from horizontal.

Cognitive checkpoint 1: slope is “same instant, different elements,” not “same element, different instants”

This is the easiest place to trip. The slope

$$\left. \frac{\partial y}{\partial x} \right|_{x,t} = \lim_{dx \rightarrow 0} \frac{y(x + dx, t) - y(x, t)}{dx}$$

is a difference between two **different elements**, x and $x + dx$, at **frozen time** t . It is *not* $y(x, t + dt) - y(x, t)$ (that’s velocity, on the $F = ma$ side).

The counter-intuition: dx is not “walking over there,” it’s “looking at the piece next door.” The string exists **whole and simultaneous**; freezing t is taking a snapshot, and on that snapshot the displacements at x and $x + dx$ are both there at the same instant — you don’t move to see the second element. x is the independent variable (a coordinate label for which piece, not moving in time); what changes with time is $y(x, t)$. The slope changes with t because “the slope field evolves in time,” not because “ x is moving.”

In one line: **same time, different elements** $\rightarrow$ **slope**; **same element, different times** $\rightarrow$ **velocity**.

Cognitive checkpoint 2: why tension pulls “outward at both ends, in opposite directions”

It feels wrong at first: both ends pull outward in opposite directions — don’t they cancel? First, why “two hands pulling apart” rather than “one hand whipping”: cut the string at x into a left piece and a right piece; at the interface the two pieces act on each other. By **Newton’s third law**, the right piece’s force on the left equals and opposes the left’s force on the right — “opposite directions” is already a theorem here, not an assumption. Now the orientation: the string is flexible, it can only **pull, not push** (push a rope and it buckles), and pulling = the two sides attract = each drags the other toward itself $\rightarrow$ both forces point outward, away from each other.

“One hand whipping” = one free end, which is a **boundary** (endpoint), not an interior element. The element $[x, x + dx]$ we’re analyzing is in the **interior**, string on both sides, so necessarily “two hands.” A hand driving one end is a boundary condition, a separate matter.

And the “cancellation” worry? **The horizontal components do nearly cancel** (this is exactly $F_x \approx 0$, A4’s tab); but **the vertical components do not**, because the two ends have different tangent angles. The net force lives in the *angle difference*, not in a magnitude difference.

So the net forces:

$$F_y = T \sin \theta(x + dx) - T \sin \theta(x) \quad (\text{vertical, drives the disturbance})$$

$$F_x = T \cos \theta(x + dx) - T \cos \theta(x) \quad (\text{horizontal, should be 0 by A2})$$

How does energy get transmitted? With $F_x = 0$ and elements not moving along x , energy still travels rightward — via the **transverse** force doing work: power = $F_y v_y$, both on y . The left element drags the right one upward, energy relays along x . “The wave travels along x ” refers to the **pattern/phase** (phase speed c), not the elements. Elements only move up/down; no mass moves along x , so $v_x = 0$ is consistent.

1.3 Newton’s second law → a differential equation

Element mass = μdx , acceleration = $\partial^2 y / \partial t^2$ (with $v_y = \partial y / \partial t$, $a = \partial v_y / \partial t = \partial^2 y / \partial t^2$; two $\partial / \partial t$ stacked become $\partial^2 / \partial t^2$, a notation, not a cancellation). $F_y = ma$:

$$T \sin \theta(x + dx) - T \sin \theta(x) = \mu dx \frac{\partial^2 y}{\partial t^2}$$

The left side is the difference of one function $g(x) = T \sin \theta(x)$ at $x + dx$ and x . Divide by dx , let $dx \rightarrow 0$:

$$\frac{T \sin \theta(x + dx) - T \sin \theta(x)}{dx} \xrightarrow{dx \rightarrow 0} \frac{\partial}{\partial x} (T \sin \theta)$$

giving

$$\frac{\partial}{\partial x} (T \sin \theta) = \mu \frac{\partial^2 y}{\partial t^2}$$

1.4 Small-angle approximation → the wave equation

Geometric bridge: $\tan \theta = \partial y / \partial x$. A3 says θ is small, and for small angles $\sin \theta \approx \theta \approx \tan \theta$:

$$\sin \theta \approx \theta \approx \tan \theta = \frac{\partial y}{\partial x}$$

Substitute:

$$\frac{\partial}{\partial x} \left(T \frac{\partial y}{\partial x} \right) = \mu \frac{\partial^2 y}{\partial t^2}$$

Pull out T (constant, A4), let $c^2 = T / \mu$:

$$\boxed{\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}}$$

1.5 The identity of c : dimensional analysis, not a coincidence

$c^2 = T/\mu$, with dimensions

$$\sqrt{\frac{T}{\mu}} = \sqrt{\frac{\text{N}}{\text{kg/m}}} = \text{m/s}$$

a speed. **This is not a coincidence, it's forced.** For a string you have exactly two quantities to work with: tension T ($\text{kg}\cdot\text{m/s}^2$) and linear density μ (kg/m). To build a speed (m/s), the **only** combination is $\sqrt{T/\mu}$ — try anything else ($T\mu$, T/μ^2 , $\sqrt{T\mu}$) and the dimensions are wrong. So that coefficient in the wave equation can't be anything else; it can only look like T/μ .

This is the power of dimensional analysis: often, before solving anything, just “what quantities do I have, what dimension do I need to build” locks the *shape* of the answer, up to a dimensionless constant. It also runs backward as a check — derive any equation, and if a coefficient's dimensions can't be assembled into what they should be, you know immediately you've erred.

1.6 Settling A4's tab

A4 (T everywhere equal) was pulled out as a constant, but it's really a consequence of $F_x = 0$: $F_x = \partial_x(T \cos \theta) dx$, and for small angles $\cos \theta \approx 1$, so $F_x \approx \partial_x T dx$; requiring $F_x = 0$ (A2, elements don't move along x) forces $\partial_x T = 0$, i.e. T constant along the string. Self-consistent, tab settled.

2. Separation of variables: pulling apart the entangled x, t

The wave equation is a PDE, x and t tangled, not directly solvable. So ask a lazy question: **is there some especially simple solution where x and t separate?**

The most naive attempt — a solution whose **spatial shape is fixed and only oscillates in overall amplitude with time**:

$$y(x, t) = a(t) f(x)$$

- $f(x)$ = the **spatial shape**: at some instant, looking along the string, the relative heights at each x (it carries its own signs: $\sin x$ is positive on the left half, negative on the right; $f = 0$ is a node).
- $a(t)$ = the **overall amplitude**: a scalar with no x , the same value across the whole string, deciding only how large the shape is scaled right now and which way.

The physical picture is a standing wave: shape frozen, breathing up and down as a whole, nodes fixed. This is not the general solution, it's the simplest class we single out first — but because the equation is **linear**, superposing this class reconstructs the general wave, so it's the “building block that reconstructs all solutions.”

Substitute into $\partial_{tt}y = c^2 \partial_{xx}y$:

$$\ddot{a}(t) f(x) = c^2 a(t) f''(x)$$

Divide by $a(t)f(x)$:

$$\frac{\ddot{a}(t)}{a(t)} = c^2 \frac{f''(x)}{f(x)}$$

Left side has only t , right side has only x — separated.

Cognitive checkpoint 3: why both sides “must equal a constant”

Write the left as $L(t)$, the right as $R(x)$; $L(t) = R(x)$ holds for **all** (x, t) .

- Fix one particular $x = x_0$: then $R(x_0)$ is a pinned-down number, so $L(t) = R(x_0)$ for all t → $L(t)$ is identically that number → L **does not vary with t** .
- Fix one $t = t_0$: similarly R **does not vary with x** .

The mechanism in one line: **the right side has no t , so the left side is not allowed to move with t ; the left side has no x , so the right side is not allowed to move with x** . Two locks clamp it: this common value varies with neither t nor x ; this system has no independent variable besides x, t , so it varies with nothing — **it can only be a constant**.

This constant, ranging over its possible values, carves out one mode per value — the origin of the $\sum_k$ to come.

Write the constant as $-k^2$ (the negative square is deliberate: only then is the solution the oscillatory $\sin / \cos$ rather than a blowing-up exponential):

$$\frac{\ddot{a}}{a} = c^2 \frac{f''}{f} = -c^2 k^2$$

Split into two independent harmonic-oscillator equations:

$$f''(x) = -k^2 f(x) \quad (\text{space})$$

$$\ddot{a}(t) = -c^2 k^2 a(t) = -\omega^2 a(t), \quad \omega \equiv ck \quad (\text{time})$$

Historical honesty: k = “wavenumber,” $\omega = ck$ = “dispersion relation,” “05 swaps the dispersion relation” — these names are **coordinates back-filled by later generations**, not the order of discovery. Real history: d’Alembert (1747) solved the wave equation first (traveling-wave form, no k, ω); Bernoulli argued for sinusoidal superposition and was doubted; Fourier (1807) gave the generality of sine series for heat conduction, and was still deemed non-rigorous by the mathematicians of his day; k, ω as terms crystallized only after 19th-century optics/acoustics matured. Our **derivation itself** is forward (we push it out step by step); the **names and “signature” readings** are a rear-view mirror — used because for someone who already knows calculus, handing over the coordinates first is more efficient, not because that’s how it was discovered.

3. Interlude: how to solve $f'' = -k^2 f$ *without* guessing e^{rx}

The textbook solves $f'' = -k^2 f$ by “assume $f = e^{rx}$, substitute to get $r = \pm ik$, use Euler’s formula to get $\sin / \cos$.” Efficient, but it can’t answer “if I didn’t know the e^{rx} route, what would I do?”

The path below uses only **Taylor expansion, matrix multiplication, series, and limits** — it does not presuppose the form of the solution, and lets the trigonometric functions be “squeezed out” of a matrix power. (This is a cleanup of a real exploration I did together with DeepSeek.)

3.1 Discretize: cut the continuous into small steps

We don’t know what f looks like, but we know Taylor expansion. For small step h :

$$\begin{aligned}f(x+h) &\approx f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 \\f'(x+h) &\approx f'(x) + f''(x)h + \frac{1}{2}f'''(x)h^2\end{aligned}$$

Substitute $f'' = -k^2 f$:

$$\begin{aligned}f(x+h) &\approx f(x) + f'(x)h - \frac{1}{2}k^2 f(x)h^2 \\f'(x+h) &\approx f'(x) - k^2 f(x)h - \frac{1}{2}k^2 f'(x)h^2\end{aligned}$$

3.2 Matrixify: pack “a pair of numbers” together

f and f' appear paired, and the transition is linear. Pack them into a state vector $F(x) = \begin{bmatrix} f(x) \\ f'(x) \end{bmatrix}$; then

$$F(x+h) \approx M F(x), \quad M = \begin{bmatrix} 1 - \frac{1}{2}k^2 h^2 & h \\ -k^2 h & 1 - \frac{1}{2}k^2 h^2 \end{bmatrix}$$

“Take one small step” = a matrix times a vector.

3.3 Powers of the matrix: find the multiplication rule

Split $M = A I + N$, with $A = 1 - \frac{1}{2}k^2 h^2$ and

$$N = \begin{bmatrix} 0 & h \\ -k^2 h & 0 \end{bmatrix}$$

The key step:

$$N^2 = \begin{bmatrix} -k^2 h^2 & 0 \\ 0 & -k^2 h^2 \end{bmatrix} = -k^2 h^2 I$$

N^2 is a scalar matrix, so N commutes with everything, and $M^n = (AI + N)^n$ expands by the binomial theorem. Separating N^j by parity ($N^{2m} = (-1)^m (k^2 h^2)^m I$, $N^{2m+1} = (-1)^m (k^2 h^2)^m N$) gives

$$M^n = \begin{bmatrix} C_n & S_n \\ -k^2 S_n & C_n \end{bmatrix}$$

with C_n, S_n two scalar series (sums of binomial coefficient $\times$ power of $A \times (-1)^m$).

3.4 Take the limit: trigonometry emerges

Walk n steps to $x = nh$. Let $h \rightarrow 0$, $n \rightarrow \infty$, $nh = x$ fixed. Using $\binom{n}{2m} \approx \frac{x^{2m}}{(2m)! h^{2m}}$ and $A^n \rightarrow 1$:

$$C_n \rightarrow \sum_{m=0}^{\infty} \frac{(-1)^m (kx)^{2m}}{(2m)!} = \cos(kx)$$

$$S_n \rightarrow \sum_{m=0}^{\infty} \frac{(-1)^m (kx)^{2m+1}}{(2m+1)! k} = \frac{\sin(kx)}{k}$$

So

$$\begin{bmatrix} f(x) \\ f'(x) \end{bmatrix} = \begin{bmatrix} \cos kx & \frac{\sin kx}{k} \\ -k \sin kx & \cos kx \end{bmatrix} \begin{bmatrix} f(0) \\ f'(0) \end{bmatrix}$$

$$f(x) = f(0) \cos(kx) + \frac{f'(0)}{k} \sin(kx)$$

The trigonometric functions emerge naturally from the limit of the discrete recursion, without assuming at any step that “the solution is trigonometric.”

3.5 Honest critique: what this route is really worth, and its debts

The most valuable thing here is **not** the selling point “no assuming e^{rx} ”; it’s that it turns the solution of $f'' = -k^2 f$ into a **special case of the matrix exponential** e^{xL} — that closing $\lim_{n \rightarrow \infty} (I + \frac{x}{n} L)^n = e^{xL}$ is the finishing stroke, reducing a second-order ODE to a first-order system $F' = LF$ with solution e^{xL} , which is the interface to every “linear evolution” downstream (Schrödinger, the growth equation, transfer matrices).

But three things must be marked honestly:

1. **The claim “more fundamental than the characteristic-equation method” doesn’t hold up.** What we did **is** the characteristic-equation method, just via a detour: M^n splitting into C_n, S_n is, at root, the eigenvalues $\pm ik$ of L — the same thing as $r = \pm ik$. e^{rx} wasn’t avoided, it was hidden inside the limit of the matrix power, and reappeared at the end as e^{xL} . The honest statement is: “the e^{rx} ansatz isn’t handed down from the sky — it’s the scalar projection of a matrix exponential.”
2. $N^2 = -k^2 h^2 I$ **is actually rediscovering the complex numbers.** The two-dimensional algebra $\{aI + bN\}$ is isomorphic to $\mathbb{C}$; $N^2 = -(\cdot)I$ is $i^2 = -1$. This is the heart of the whole thing — and it has a beautiful

corollary: flip the sign to $f'' = +k^2 f$ and then $N^2 = +k^2 h^2 I$, $\{I, N\}$ is isomorphic to the **split-complex numbers** ($j^2 = +1$), and the limit naturally gives $\cosh, \sinh$. **The real punchline: the sign of the ODE determines the number system behind it (complex vs. split-complex), and trig/hyperbolic are just the same matrix-exponential machine projected into two different number systems.**

3. **Convergence is hand-waved (an IOU).** §3.4 swaps “limit of each term” with “sum over infinitely many terms” without proving uniform convergence; the account of local truncation error $O(h^3)$, over $n = x/h$ steps, vanishing in the limit, is also not closed. Irrelevant to a blog reader, but recorded as a debt.

4. Boundary conditions → mode discretization

The single-mode solution now carries two free constants, and k is still arbitrary and continuous. **Boundary conditions** sieve it into a discrete series.

Both ends fixed: $y(0, t) = 0, y(L, t) = 0$.

- $f(0) = 0$: $f(0) \cos 0 + \frac{f'(0)}{k} \sin 0 = f(0) = 0$, so the $\cos$ term is gone entirely, leaving $f(x) = \frac{f'(0)}{k} \sin(kx)$. ($\cos$ is nonzero at $x = 0$ and is sieved out by this boundary.)
- $f(L) = 0$: $\sin(kL) = 0 \rightarrow kL = n\pi$ (n integer, **not** $2\pi n$ — $\sin$ is zero at $\pi, 2\pi, 3\pi, \dots$):

$$k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

- $n = 0$: $k = 0, f = \sin 0 = 0$, string flat, no mode → discard.
- $-n$: $\sin(-k_n x) = -\sin(k_n x)$, differing only by an overall sign, absorbed into the amplitude coefficient, the same mode as $+n \rightarrow$ keep only $n > 0$.

This one physical constraint, “both ends fixed,” chops the infinite continuum of k into a discrete series of standing waves — the n in the $\sum_n$ of a Fourier series comes from exactly this, it’s not hard-coded.

Change the boundary, and the basis changes (same operator $-d^2/dx^2$; change the boundary and the eigenfunctions switch to a different set): - Both ends free (Neumann) → $\{\cos(k_n x)\}$ - Periodic → both $\{\sin, \cos\}$ survive (equivalently $\{e^{ikx}\}$) = the standard Fourier series

Our string is fixed at both ends, so the basis is the pure $\{\sin(k_n x)\}$. **Everything below stays in this real basis — no sneaking into the e^{ikx} complex basis or into a continuous/infinite domain.**

5. Complete orthogonal basis → general solution

5.1 Single mode → superposition

Each n gets a mode solution:

$$y_n(x, t) = \sin(k_n x) [a_n \cos(\omega_n t) + b_n \sin(\omega_n t)], \quad \omega_n = ck_n$$

Any solution = a superposition of these modes:

$$y(x, t) = \sum_{n=1}^{\infty} \sin(k_n x) [a_n \cos \omega_n t + b_n \sin \omega_n t] = \sum_{n=1}^{\infty} c_n(t) \sin(k_n x)$$

with $c_n(t) = a_n \cos \omega_n t + b_n \sin \omega_n t$. **The basis $\sin(k_n x)$ handles only x ; all of the time goes into the coefficient $c_n(t)$.**

This step relies on **two things** together:

- **Linearity + superposition** → each y_n is a solution, and superposing is still a solution → this gives that the sum **is** a solution (the “solution” part).
- **Completeness** → guarantees that **all** solutions are covered by this sum, none escaping → this gives that it is the **general** solution (the “general” part).

5.2 A function is a vector in an infinite-dimensional space (the $\mathbb{R}^3$ analogy)

This is the core of the whole block (and of 06, Hilbert space). Expanding any wave in a basis is **structurally isomorphic** to coordinate decomposition in 3D space:

$\mathbb{R}^3$	String (function space)
Basis vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$ (3)	Basis shapes $\sin(k_n x)$ (countably infinite)
Coordinates v_i	Coefficients c_n
Dot product $\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i$	Inner product $\langle f, g \rangle = \int_0^L f g dx$
Orthogonal $\hat{e}_i \cdot \hat{e}_j = 0$	$\int_0^L \sin(k_m x) \sin(k_n x) dx = 0$ ($m \neq n$)
Point $\mathbf{v} = \sum v_i \hat{e}_i$	Shape $f(x) = \sum a_n \sin(k_n x)$
Dimension 3	Dimension ∞

Not just a metaphor: in $\mathbb{R}^3$ orthogonality lets you project out a coordinate $v_i = \frac{\mathbf{v} \cdot \hat{e}_i}{\hat{e}_i \cdot \hat{e}_i}$; in function space, likewise

$c_n = \frac{\langle f, \sin k_n x \rangle}{\langle \sin k_n x, \sin k_n x \rangle}$. The “vector = coordinate $\times$ basis” geometry transplants verbatim onto functions, with the dot product replaced by an integral and 3 axes by infinitely many shapes. **A function is a vector in an infinite-dimensional space.**

5.3 Orthogonality → projection → unique coefficients

Extract the coefficient c_m : multiply both sides by $\sin(k_m x)$ and integrate over $[0, L]$; orthogonality leaves only the $n = m$ term alive:

$$\int_0^L \psi(x) \sin(k_m x) dx = \sum_n c_n \int_0^L \sin(k_n x) \sin(k_m x) dx = c_m \cdot \frac{L}{2}$$

$$c_m = \frac{2}{L} \int_0^L \psi(x) \sin(k_m x) dx$$

Uniqueness of coefficients is a chain of theorems, not hand-waving:

orthogonal $\Rightarrow$ linearly independent $\Rightarrow$ coefficients unique $\Rightarrow$ term-by-term equality

(Linear independence: suppose $\sum_n \gamma_n \sin(k_n x) = 0$; multiply by $\sin(k_m x)$, integrate, orthogonality leaves only $n = m \rightarrow \gamma_m = 0$ for every m .)

The time factor $c_n(t)$ arises from **two independent sources meeting and being self-consistent**: projection (space) gives $c_n(t) = \frac{2}{L} \int_0^L y(x, t) \sin(k_n x) dx$; dynamics (time) substitutes $y = \sum c_n(t) \sin(k_n x)$ back into the wave equation and, by term-by-term equality, gives $\ddot{c}_n = -\omega_n^2 c_n$, solving to $c_n(t) = a_n \cos \omega_n t + b_n \sin \omega_n t$. Substituting the latter into the former and verifying via orthogonality, the two are consistent.

IOU: the above repeatedly used the orthogonality integral $\int_0^L \sin(k_m x) \sin(k_n x) dx = \frac{L}{2} \delta_{mn}$ (obtainable by product-to-sum + integration; it is the shared foundation of expansion, linear independence, uniqueness, and the self-consistency check). We **skip the hand computation and record a debt**.

5.4 Completeness: free in finite dimensions, a hard theorem in infinite dimensions

In $\mathbb{R}^3$ completeness is free: you already know the space is 3D, count 3 orthogonal vectors, $3 = 3$ spans automatically. **Finite-dimensional: number of orthogonal vectors = dimension $\Rightarrow$ complete.**

Function space is infinite-dimensional, and this freebie **vanishes**: you have countably infinitely many orthogonal $\sin(k_n x)$, but “infinitely many orthogonal vectors” does **not** guarantee spanning (counterexample: take only even- n sin; still pairwise orthogonal, yet missing all odd modes). So:

- **Orthogonality**: $\int \sin k_m x \sin k_n x = 0$, direct integration, **easy**.
- **Completeness** (no nonzero shape is perpendicular to all $\sin k_n x$): **hard**, the core of the Fourier-series convergence theorem.

Completeness, formalized ($L^2[0, L]$, inner product $\langle f, g \rangle = \int_0^L f g$, basis $\phi_n = \sin k_n x$), the following are equivalent:

1. **No leakage**: $\langle f, \phi_n \rangle = 0 \forall n \Rightarrow f = 0$ (in the L^2 sense, $\|f\| = 0$).
2. **Expandable**: $\|f - \sum_{n=1}^N a_n \phi_n\| \xrightarrow{N \rightarrow \infty} 0$ (L^2 mean-square convergence, not pointwise).
3. **Parseval**: $\|f\|^2 = \sum_n a_n^2 \|\phi_n\|^2$.

Its true origin: $\{\sin k_n x\}$ are **all** the eigenfunctions of the self-adjoint operator $-\frac{d^2}{dx^2}$ (Dirichlet boundary), and by the **Sturm–Liouville completeness theorem** they span L^2 ; the rigorous proof relies on the spectral theorem for compact self-adjoint operators (functional analysis).

IOU + methodology: we **black-box completeness, cite Sturm–Liouville, and do not descend**.

See the “descent criterion” at the end — completeness’s conclusion doesn’t change how we use it upstream (expanding $P(k)$, mode-decomposing in SBI use the conclusion), so black-boxing is the mature choice, not laziness. This also stops exactly at the line where engineering functional analysis stops: use Hilbert geometry + trust the completeness theorem, don’t prove down to the measure-theory layer.

5.5 What the wavenumber k actually is

Tripped over repeatedly, so nailed down: k is **neither a count of waves nor an index**; it is “**how dense**” — how fast the phase increases per unit length, equivalently “how many complete cycles fit in a length of 2π .” $\sin(kx)$ over 2π has its phase kx increase by $2\pi k$, exactly k whole cycles.

Three things not to conflate:

- k (**wavenumber**): how dense the wave is. A continuous physical quantity with units (1/length), generally not an integer ($k_1 = \pi/L$ is usually irrational).
- n (**index**): which mode. An integer label 1, 2, 3, ...
- $k_n = n\pi/L$: maps the index n to its wavenumber.

n **orders**, k **measures density**, $k_n = n\pi/L$ **connects them**. The name “wavenumber” is a literal translation of *wavenumber* (“how many waves per unit length”) — it means a density, not a count. It is the **spatial twin of frequency**: temporal frequency = cycles per second, wavenumber = cycles per unit length.

6. The uncertainty principle $\Delta x \cdot \Delta k \geq \frac{1}{2}$

6.1 Two probability distributions

Of the same wave packet you can ask two things:

- **Where it is in space**: $|\psi(x)|^2$ as a probability density on position ($\psi(x)$ is the string’s shape at some fixed instant).
- **Which wavenumbers it’s made of**: $|c_n|^2$ (discrete) / $|\tilde{\psi}(k)|^2$ (continuous) as a probability density on wavenumber.

Δx and Δk are the **standard deviations** of these two distributions.

Honesty checkpoint: “**amplitude squared = probability**” is the **Born rule (a postulate)**, not a **derivation**

Everything so far was self-provable step by step, so at $|\psi|^2$ one expects it to be derivable too — but it isn’t. This is the **seam between mathematics and physics**:

Mathematics only gave you “a bunch of normalized nonnegative weights” ($\int |\psi|^2 = 1$, from normalization + orthogonality + Parseval). It **looks like** a probability distribution (nonnegative, sums to 1), but “looks like a probability” $\neq$ “is the probability of some measurement.” Why squared, not $|\psi|$ or $|\psi|^4$? At the math level, because **Parseval conserves the square** ($\|\psi\|^2 = \sum |c_n|^2$, the Pythagorean theorem), and the square is the only power that keeps the total conserved.

The step from “looks like a probability” to “is a probability” is the **Born rule**, a **postulate** of quantum mechanics — guessed (Born added the square in a footnote in 1926), verified by experiment over and over, never overturned, hence accepted as an axiom. A few corroborations (not proofs): total-probability conservation (unitarity) requires the integral of $|\psi|^2$ conserved; probability nonnegativity is automatic since $|\psi|^2 \geq 0$; Gleason’s theorem says that for Hilbert dimension ≥ 3 a probability assignment **must** be of the form $|\langle \cdot \rangle|^2$ (but it presupposes “describe states with a Hilbert space”).

So the honest way to write “amplitude squared = probability”: **square-conservation is math (true); “= probability” is interpretation / postulate.** On the string, $|\psi|^2$ is at most an energy density with no probabilistic meaning; only in quantum mechanics, via the Born rule, is it **defined** to be measurement probability. **Record a debt, accept it as a postulate, do not descend** (there’s nothing more fundamental below to derive from — this is the bedrock).

6.1b Legitimacy of the probability-density definition: along which axis, with which measure

Above we called $|\psi(x)|^2$ a “probability density” and also called $|\tilde{\psi}(k)|^2$ a “probability density.” There’s a legitimacy issue here, easy to slide past but worth nailing: **these are densities on two different spaces, each defined relative to a measure, and they must not be conflated.**

A “density” is always relative to some measure. Saying $|\psi(x)|^2$ is a probability density means precisely: the probability that the position falls in a small interval $[x, x + dx]$ is $|\psi(x)|^2 dx$. Here dx is **Lebesgue measure on the x -axis**. In measure-theoretic terms, the probability measure $\mathbb{P}_x$ is **absolutely continuous** with respect to Lebesgue measure dx , and $|\psi(x)|^2$ is its **Radon–Nikodym derivative**:

$$d\mathbb{P}_x = |\psi(x)|^2 dx, \quad |\psi(x)|^2 = \frac{d\mathbb{P}_x}{dx}$$

Likewise $|\tilde{\psi}(k)|^2$ is the density of **another** probability measure $\mathbb{P}_k$ with respect to **Lebesgue measure dk on the k -axis**:

$$d\mathbb{P}_k = |\tilde{\psi}(k)|^2 dk, \quad |\tilde{\psi}(k)|^2 = \frac{d\mathbb{P}_k}{dk}$$

Two axes, two measures, two probability spaces. $|\psi|^2$ integrating to 1 over the x -axis and $|\tilde{\psi}|^2$ integrating to 1 over the k -axis are **two independent statements**, not the same integral in different notation. Each is legitimate on its own (each nonnegative, each normalized); and “both normalized at once” is not automatic — it’s **guaranteed by Parseval**: change of basis (position basis $\leftrightarrow$ wavenumber basis) is a unitary transform, length-preserving, so $\int |\psi|^2 dx = \int |\tilde{\psi}|^2 dk = 1$ hold together.

Cognitive checkpoint: “integrate along x vs. integrate along k ” are not the same thing

$\int (x - \bar{x})^2 |\psi(x)|^2 dx$ computes the position variance on the x -axis with x ’s measure; $\int (k - \bar{k})^2 |\tilde{\psi}(k)|^2 dk$ computes the wavenumber variance on the k -axis with k ’s measure. The same physical wave packet, projected onto two different measure spaces, has a statistic computed once in each. Δx and Δk are **the standard deviations in two different spaces** — the uncertainty principle states that the product of these “variances living on different axes” has a lower bound, which is also why it’s not a trivial statement: two spaces that were mutually irrelevant get bound into a product lower bound by the Fourier bridge.

Discrete vs. continuous: the measure changes, and so does the normalization constant. On the finite string $[0, L]$ the wavenumbers are discrete, $k_n = n\pi/L$, and the “wavenumber distribution” lives on a **counting measure**: $\sum_n (\cdots)$ rather than $\int dk$. Normalization is then $\sum_n P_n = 1$ (weights $P_n = \frac{L}{2} |c_n|^2$, where that $\frac{L}{2}$ is the normalization constant left over from the un-normalized sin basis, see §6.4). Put the string on an infinite line

($L \rightarrow \infty$), the discrete spectrum condenses into a continuum, the counting measure $\sum_n$ passes to Lebesgue measure $\int dk$, and $c_n \rightarrow \tilde{\psi}(k)$. **Same statistics, the underlying measure switching from “counting” to “Lebesgue,” the sum becoming an integral** — this is exactly the measure-theoretic landing of the “discrete $\sum_n \leftrightarrow$ continuous $\int dk$ ” remark in §5.

IOU (measure theory, black-boxed): the three pieces used above — (i) existence/uniqueness of the Radon–Nikodym derivative (absolute continuity $\Rightarrow$ a density exists), (ii) the construction of Lebesgue / counting measure, (iii) the measure-convergence of discrete spectrum to continuous as $L \rightarrow \infty$ — are all measure-theory foundations we **accept the conclusions of and do not descend into**. By the “descent criterion” below: these conclusions don’t change how we use the density upstream (computing variances, computing the bound), so black-box. All we need is the **license** “ $|\psi|^2$ is a legitimate density w.r.t. dx ,” not its issuance process. This is consistent with everything before: the completeness of L^2 , why Lebesgue integration replaces Riemann (to handle functions on the domain that are pathological — discontinuous down to the measure level — and to make limits closed) — all stop at the line where engineering functional analysis stops.

Takeaway 1: Fourier change-of-basis $\neq$ Bayesian update (a common mis-association)

$|\tilde{\psi}(k)|^2$ being a probability, with a flavor of “inferred backward from evidence,” easily suggests Bayes $p(\theta | d) \propto p(d | \theta)p(\theta)$. But that’s a mis-association. The core of Bayes is **conditioning**: given evidence d , bend the prior into a posterior — “information update.” Whereas $\psi(x) \leftrightarrow \tilde{\psi}(k)$ is **the coordinates of the same object in two orthogonal bases** (position basis vs. wavenumber basis), the core being “change of coordinates / change of representation,” not “update” — no evidence enters, no condition, nothing updating anything. In one line: **Bayes is conditioning (given d , change the distribution); Fourier is unitary change-of-basis (view the same state in a different basis)**. Both handle probabilities, common ancestor “probability + linear algebra,” but one is a **rotation** and the other a **projection + normalization**, two different operations. The real commonality you sensed is “the same object, two complementary representations, each carrying a probability distribution” — that “complementary representation” instinct is right (it’s precisely the prototype of position representation vs. momentum representation later in quantum mechanics); just don’t file “change-of-basis” as “update.”

Takeaway 2: the basis itself is not a probability; the squared components are (this is Hilbert geometry, not measure theory)

$\sin(k_n x)$ is a **basis vector** (an axis direction); it is not a probability. What carries probabilistic meaning is the **square of the coefficient** $|c_n|^2$ — “how much this state occupies along the n -th axis.” Just as in $\mathbb{R}^3$ an axis $\hat{e}_i$ isn’t a probability, but the squared components v_i^2 of a unit vector sum to 1 and can serve as “the probability of landing on each axis.” So the instinct “orthogonal decomposition $\rightarrow$ squared weights per direction $\rightarrow$ looks like a probability distribution” stands on **Hilbert geometry** (normalization + orthogonality $\Rightarrow \sum |c_n|^2 = \|\psi\|^2$, the infinite-dimensional Pythagorean theorem), **not measure theory**. Measure theory lives further down, minding a different matter: how the integral $\int |\psi|^2 dx$ is **rigorously defined**, why L^2 is complete — it lays the foundation for “integral/probability,” it doesn’t handle the “basis $\rightarrow$ probability” instinct, and only shows up when you ask “what exactly is this integral.” **Layer them: “orthogonal decomposition yields probabilities” = Hilbert geometry (Pythagoras); “why the integral/density is legitimate” = measure theory**

(foundation).

6.2 A wave packet = a sum of sines (why big $\Delta k \rightarrow$ small Δx)

Any wave packet = a large bunch of sines of different wavenumbers, added with weights. In the middle segment the component waves are **in phase** and reinforce $\rightarrow$ bulging into a packet; on the two sides they are **out of phase** and cancel $\rightarrow$ flattening to zero. This is the mechanism by which a heap of sines spread over all space superpose into a **localized** packet.

The more wavenumbers mixed in, the wider they spread (Δk large), the cleaner the cancellation on the sides and the narrower the bulge (Δx small); conversely, using only a few nearby wavenumbers (Δk small), cancellation is incomplete and the packet is wide (Δx large). One large forces the other small — this is the visual root of $\Delta x \cdot \Delta k \geq \frac{1}{2}$.

6.3 Definitions (discrete version, staying in the sin basis / $[0, L]$)

Fix t , normalized. Position distribution $|\psi(x)|^2$, $\psi(x) = \sum_n c_n \sin(k_n x)$; wavenumber distribution $|c_n|^2$ (weight of the n -th mode, $k_n = n\pi/L$):

$$\begin{aligned}\bar{x} &= \int_0^L x |\psi|^2 dx, & \Delta x^2 &= \int_0^L (x - \bar{x})^2 |\psi|^2 dx \\ \bar{k} &= \sum_n k_n |c_n|^2, & \Delta k^2 &= \sum_n (k_n - \bar{k})^2 |c_n|^2\end{aligned}$$

Variance = mean of the square – square of the mean: $\Delta x^2 = \langle x^2 \rangle - \bar{x}^2$ (expand $(x - \bar{x})^2$, use the definition of $\bar{x}$ and normalization), and Δk likewise.

Proposition: $\Delta x \cdot \Delta k \geq \frac{1}{2}$, equality $\iff$ Gaussian wave packet.

6.4 Proof (in the sin basis, no i , purely real)

Reduction. Set $\bar{x} = 0$, $\bar{k} = 0$. This **loses no generality**: translating x , and shifting $\bar{k}$ by multiplying ψ by a phase, are **translations**, and $\Delta x, \Delta k$ are translation-invariant. The case of nonzero means merely carries extra constant terms, no new content. So we must show

$$\left(\int_0^L x^2 |\psi|^2 dx \right) \left(\sum_n k_n^2 |c_n|^2 \right) \geq \frac{1}{4}$$

The bridge: move Δk^2 from wavenumber into x -space (second derivative). Each basis satisfies $-\frac{d^2}{dx^2} \sin(k_n x) = k_n^2 \sin(k_n x)$, so $-\psi'' = \sum_n k_n^2 c_n \sin(k_n x)$. Multiply by ψ , integrate, and orthogonality keeps only the diagonal terms:

$$\sum_n k_n^2 |c_n|^2 \propto \int_0^L \psi(-\psi'') dx$$

(strictly with a normalization constant $L/2$; taking the wavenumber probability weight as $P_n = \frac{L}{2}|c_n|^2$ — which is the distribution truly satisfying $\sum_n P_n = 1$ — absorbs the $L/2$ into P_n and the equality is clean). Integrate by parts, the boundary term $[\psi\psi']_0^L$ vanishing since both ends are fixed $\psi(0) = \psi(L) = 0$:

$$\int_0^L \psi(-\psi'') dx = \int_0^L |\psi'(x)|^2 dx$$

Physical meaning: Δk^2 = the total squared slope of the waveform = how violently the wave changes, matching “big wavenumber = fast change.” So

$$\Delta x^2 \Delta k^2 = \left(\int_0^L x^2 |\psi|^2 dx \right) \left(\int_0^L |\psi'|^2 dx \right) = \|x\psi\|^2 \|\psi'\|^2$$

Cauchy–Schwarz ($f = x\psi$, $g = \psi'$; ψ real, conjugate omitted):

$$\|x\psi\|^2 \|\psi'\|^2 \geq \left| \int_0^L x\psi\psi' dx \right|^2$$

Integration by parts pins down the right side: $x\psi\psi' = \frac{1}{2}x \frac{d}{dx}(\psi^2)$,

$$\int_0^L x\psi\psi' dx = \frac{1}{2}[x\psi^2]_0^L - \frac{1}{2} \int_0^L \psi^2 dx = -\frac{1}{2}$$

(boundary term: $\psi(L) = 0$ and $x = 0$ at the other end, so it vanishes; the last step uses normalization $\int_0^L \psi^2 = 1$). Therefore

$$\Delta x^2 \Delta k^2 \geq \left| -\frac{1}{2} \right|^2 = \frac{1}{4} \implies \boxed{\Delta x \cdot \Delta k \geq \frac{1}{2}}$$

Equality needs Cauchy–Schwarz to be tight, $\psi' = \lambda x\psi$, solving to $\psi \propto e^{\lambda x^2/2}$, and normalizability forces $\lambda < 0$ → **Gaussian wave packet.** ■

6.5 Connecting to Heisenberg

$\Delta x \cdot \Delta k \geq \frac{1}{2}$ is a **pure Fourier fact** — one object expanded in a “position basis” and a “wavenumber basis,” the product of spreads having a lower bound. It is **independent of the specific basis chosen** (provable in $\sin / \cos / e^{ikx}$ alike); at bottom it’s about “two mutually Fourier-dual bases.”

05 adds **one physical input**, $p = \hbar k$ (momentum = $\hbar \times$ wavenumber), and multiplying through:

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

— **Heisenberg’s uncertainty principle** = **this Fourier inequality** + “ $p = \hbar k$.” Those “two physical quantities” are position x and momentum p , which are **a pair of mutually Fourier-transformed bases** (position representation $\leftrightarrow$ momentum representation) — the product of spreads of a *dual pair* has a lower bound, not any two quantities.

7. The debts of this block (the IOU ledger)

Honestly listing the black boxes, rather than pretending they’re settled:

1. **The orthogonality integral** $\int_0^L \sin(k_m x) \sin(k_n x) dx = \frac{L}{2} \delta_{mn}$ — obtainable by product-to-sum + integration, hand computation skipped. It is the shared foundation of expansion / linear independence / uniqueness / self-consistency check.
2. **Completeness** of $\{\sin k_n x\}$ spanning $L^2[0, L]$ — cite Sturm–Liouville; rigorous proof relies on the compact self-adjoint spectral theorem, black-boxed.
3. **The Born rule** “ $|\psi|^2$ = probability” — a physical postulate, not a derivation; math only reaches “normalized nonnegative weights.” Accepted, not descended into.
4. **The continuous Fourier transform** $\psi(x) \leftrightarrow \tilde{\psi}(k)$ — this whole post stays on the finite string, sin basis / $[0, L]$ / discrete; it never establishes the passage “discrete $\sum_n \rightarrow$ continuous $\int dk$ ” ($L \rightarrow \infty$). The continuous version is something to be established separately.
5. **Convergence details of the ODE limit** (§3) — the interchange of limit and sum, and the account from local $O(h^3)$ to global convergence, hand-waved.
6. **The measure-theoretic foundation of probability density** (§6.1b) — existence/uniqueness of the Radon–Nikodym derivative, construction of Lebesgue / counting measure, the measure-convergence of discrete to continuous spectrum as $L \rightarrow \infty$, and why Lebesgue integration replaces Riemann (pathological functions + L^2 completeness). We accept the license “ $|\psi|^2$ is a legitimate density w.r.t. dx ” and don’t descend to its issuance.

These debts **do not damage first principles** — first principles is “knowing why each step holds and honestly marking the black boxes,” not “welding the rebar of every foundation slab.”

8. Methodology: when to black-box, when to descend

Repeatedly during the derivation we hit “should I dig into the proof of some prerequisite.” The criterion, in one line:

Will this proof’s conclusion change how I use it upstream? - **No** $\rightarrow$ black-box it, record a debt, move up. - **Yes** $\rightarrow$ then it’s worth descending.

Against this post:

- **The completeness theorem:** its conclusion (“expandable into $\sum$ ”) is used without touching its proof $\rightarrow$ black-box.
- $N^2 = -I \Rightarrow \mathbb{C}$: it changes how you see the whole machine (trig/hyperbolic are the same matrix exponential projected into two number systems) $\rightarrow$ worth deriving by hand.

The termination condition = **the engineering-physics floor + recording rigor as IOUs**, preventing infinite recursive descent (functional analysis $\rightarrow$ real analysis $\rightarrow$ measure theory $\rightarrow$ topology... is bottomless). For the measure theory / Hilbert space / completeness / convergence bundle: **accepting the abstract conclusions is enough for applications; when you genuinely need theoretical innovation someday, dig then.** Stopping at the line where engineering functional analysis stops is maturity, not a shortfall — even researchers trust in layers; nobody proves every theorem they use down to the bottom, time doesn't physically allow it. What distinguishes the strong isn't "did you prove it all the way down," it's **knowing which pieces you must chew yourself and which you can trust.**

9. This block's results and interfaces

Results: wave equation $\rightarrow$ separation of variables / modes / complete superposition $\rightarrow \Delta x \cdot \Delta k \geq \frac{1}{2}$.

Interfaces (straight into what follows):

- $\Delta x \cdot \Delta k \geq \frac{1}{2} \rightarrow 05$ adds $p = \hbar k$, i.e. Heisenberg.
- "complete-basis superposition + coefficients evolving in t " $\rightarrow$ the prototype of the Hilbert-space state vector in 06, and the same machinery for mode-decomposing the density field and $P(k)$ in later cosmology.

Next block, 05 • The Schrödinger equation: the core move is "swap in a different dispersion relation" — the string has $\omega = ck$, quantum swaps in $\omega \propto k^2$; there an i will appear, hammered into the equation, and the §3 viewpoint $N^2 = -I \Rightarrow \mathbb{C}$ is exactly what lets you ask "by what right this i ."